ShiftLens: From Controlled Shifts to Evidence-Gated Attribution

A reproducible two-factor protocol for cross-corpus hate-speech evaluation, uncertainty-aware attribution, and claim discipline

Corpus Intake & Reliability

- Anchor corpus: YouTube-BN
- Targets: BD-SHS, HateXplain
- BengaliHate v1/v2 excluded when labels are unreliable
- Provenance and label-quality checks required

Leakage-Safe Data Protocol

- Normalize text and label schemas
- Audit duplicates and cross-corpus overlap
- Five disjoint pools: train / val / adapt / test / reserve
- Class prior and sample size preserved

Controlled Interventions

- Content: class-preserving reserve-pool mixture
- Protocol: declared transition matrix T
- Baseline, content-only, protocol-only, joint
- 2 × 2 factorial, 5 independent seeds

Model Panel & Completeness

- Majority, TF-IDF + Logistic Regression
- ERM (XLM-R), DANN, CORAL, protocol-corrected
- Source-only validation checkpoints
- 6 × 4 × 5 = 120 expected runs

Evaluation & Attribution

- Macro-F1, AUROC, AUPRC, ECE
- Exact Shapley: main effects + interaction
- Paired randomization, seed intervals
- Robustness grid + external stress test

Evidence & Claim Gate

- Requires complete factorial coverage
- Reports multi-seed uncertainty
- Separates diagnostic evidence from causal claims
- Blocks unsupported headline conclusions